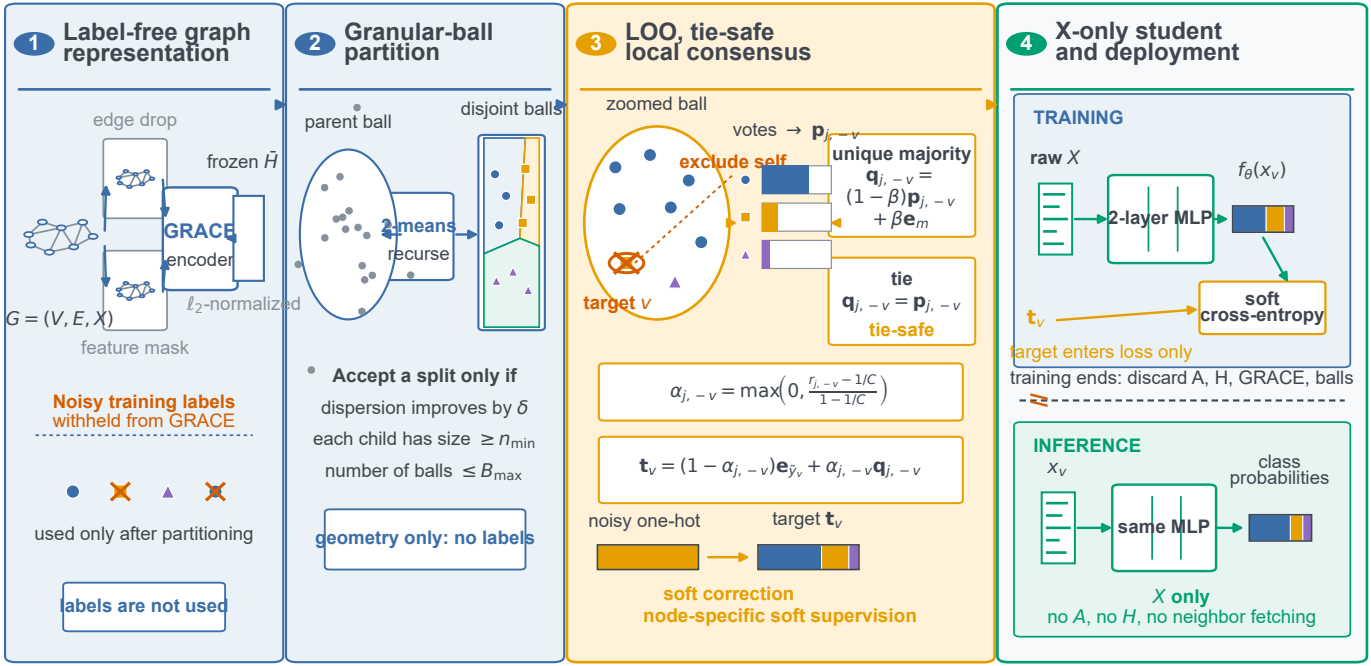

1
Label-free graph representation
edge drop
frozen $\bar{H}$
GRACE
encoder
ℓ_2-normalized
$G = (V, E, X)$
feature mask
Noisy training labels
withheld from GRACE
used only after partitioning
labels are not used
2
Granular-ball partition
parent ball
disjoint balls
2-means
recurse
Accept a split only if
dispersion improves by δ
each child has size $\geq n_{\min}$
number of balls $\leq B_{\max}$
geometry only: no labels
3
LOO, tie-safe local consensus
zoomed ball
votes $\rightarrow$ $\mathbf{p}_{j,-v}$
exclude self
target v
unique majority
$\mathbf{q}_{j,-v} = (1-\beta)\mathbf{p}_{j,-v} + \beta\mathbf{e}_m$
tie
$\mathbf{q}_{j,-v} = \mathbf{p}_{j,-v}$
tie-safe
$\alpha_{j,-v} = \max\left(0, \frac{r_{j,-v} - 1/C}{1 - 1/C}\right)$
$\mathbf{t}_v = (1 - \alpha_{j,-v})\mathbf{e}_{\tilde{y}_v} + \alpha_{j,-v}\mathbf{q}_{j,-v}$
noisy one-hot
target $\mathbf{t}_v$
soft correction
node-specific soft supervision
4
X-only student and deployment
TRAINING
raw X
2-layer MLP
$f_\theta(x_v)$
$\mathbf{t}_v$
soft cross-entropy
target enters loss only
training ends: discard A, H, GRACE, balls
INFERENCE
x_v
same MLP
class probabilities
X only
no A, no H, no neighbor fetching